

Key Terms

- Algorithms
- Assembler
- Assembly Language
- Bits and Bytes
- Bytecode
- Case-Sensitive Language
- Compiler
- Computer Programs
- Computer Programming
- High Level Language
- IDE (Integrated Development Environment)
- Interpreter
- Logic error
- Machine Language
- Memory and address
- Object-Oriented Programming
- PI (Python Interpreter)
- Procedural / Structured Programming
- Pseudo Code
- PVM (Python Virtual Machine)
- Runtime error
- SDK (Software Development Kit)
- Syntax error

Review Questions

Fill in the blanks in each of the following sentences about the Python environment.

1. A _____ is a binary digit 0 or 1.
2. A _____ is a sequence of 8 bits.
3. A _____ translates the entire source code into a machine code file, which is then executed.
4. An _____ reads one statement from the source code, translates it to machine code or virtual machine code, and then executes it immediately.
5. The _____ command from the Software Development Kit executes a Python application.
6. A Python program file must end with the _____ file extension.
7. Programming errors can be categorized into three types: _____, _____, and _____.

Answer: 1. bit, 2. byte, 3. compiler, 4. interpreter, 5. python, 6. .py, 7. syntax errors, runtime errors, logic errors.

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Key Terms

- Boolean
- Case-sensitive language
- Comments
- Data types
- Floating point
- Indentation 缩进
- Identifiers 变量、类……名称
- Immutable 不可更改
- Initialization
- Integer
- Literals 保存在变量中的数
- Mutable
- Named constants
- Objects
- Reserved words
- Snake case
- String
- Type conversion
- Type hinting
- Variables

Review Questions

Fill in the blanks in each of the following sentences about the Python environment.

1. Comments in programs are for human to read, and to increase their readability. They are added using hash symbols and triple quotes.
2. main() is the entry point of a Python program/script.
3. objects are used to store data in a program. They are created by being assigned a value, and should be referenced using a variable.
4. An identifier is the name used for an element in a program. It is a sequence of characters of any length consists of letters, digits, and underscore. It must start with a letter or an underscore. It cannot start with a digit. It cannot be a reserved word.
5. Literals are values associated with data types.

Answer: 1. Comments; hash symbols; triple quotes, 2. main(), 3. objects; value; variable, 4. identifier; letters; digits; underscores; letter; underscore; digit; reserved word, 5. Literals.

Review Questions

Fill in the blanks in each of the following sentences about the Python environment.

6. It is a good practice to specify the type of a variable using a _____. We call it _____.
7. A _____ represents permanent data that never changes, and its variable uses all _____ letters.
8. You can output data to the standard output device using the _____ function. The parameter _____ is used to specify the separator, and _____ is used to specify the ending character.
9. You can get input using the _____ function and convert a string into a numerical value using the _____ function or the _____ function.
10. In Python, variable names with more than one word are separated using _____, and we call it _____.
11. The difference in _____ disturbs Python's ability to translate the source code.

Answer: 6. colon, type hinting, 7. named constant, capital, 8. print, sep, end, 9. input, int, float, 10. underscores, snake case, 11. indentation.

Review Questions

Fill in the blanks in each of the following sentences about the Python environment.

6. It is a good practice to specify the type of a variable using a colon. We call it type hinting.
7. A named constant represents permanent data that never changes, and its variable uses all capital letters.
8. You can output data to the standard output device using the print function. The parameter sep is used to specify the separator, and end is used to specify the ending character.
9. You can get input using the input function and convert a string into a numerical value using the int function or the float function.
10. In Python, variable names with more than one word are separated using underscores, and we call it snake case.
11. The difference in indentation disturbs Python's ability to translate the source code.

Answer: 6. colon, type hinting, 7. named constant, capital, 8. print, sep, end, 9. input, int, float, 10. underscores, snake case, 11. indentation.